

NEURIPS 2025

SmoIVLM

Redefining Small and Efficient Multimodal Models

Hugging Face & Stanford University

arXiv:2504.05299

256M

Parameters

<1GB

GPU RAM

80

Tokens/sec



or scroll to navigate

Large VLMs Are Powerful but Impractical for Edge Deployment

State-of-the-Art VLM RAM

27.7 GB

Molmo-A1B-7B requires massive GPU memory

Edge Device Constraint

~4-8 GB

Smartphones and laptops are memory-limited

Scale Gap

300×

Idedics-80B vs SmoIVLM-256M performance

SmoIVLM Solution

0.8 GB

Sub-1GB inference enables edge deployment

SmolVLM: Image Splitting, Pixel Shuffle, and SmolLM2 Backbone

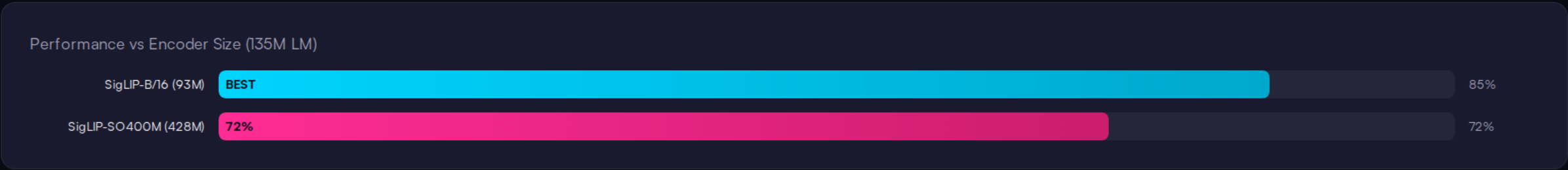


► Images split into sub-images for efficient encoding at varying resolutions

► Pixel shuffle (space-to-depth) compresses spatial features into channels

► Text tokens concatenated with visual features for multimodal input

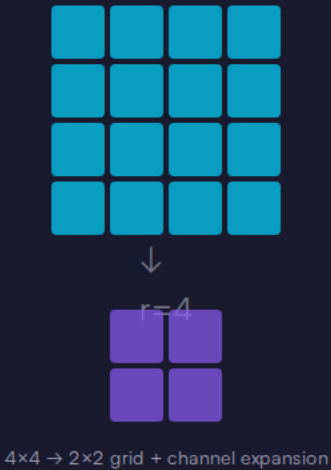
Smaller Vision Encoders Outperform Larger Ones in Compact VLMs



Key Insight: A 93M encoder outperforms a 428M encoder with the same 135M language model. The larger encoder's 10% parameter overhead is only justified at 1.7B+ LM scale. Balanced encoder-LM compute allocation is critical for compact VLMs.

Aggressive Pixel Shuffle (r=4) Cuts Visual Tokens 16× for Small Models

Pixel Shuffle: r=4 (16× Reduction)



Compression Factor Selection

Model Size	Optimal r	Token Reduction
256M / 500M	r = 4	16× fewer tokens
2.2B	r = 2	4× fewer tokens

Small VLMs benefit from r=4: reduces attention overhead, improves long-context modeling.

Extending Context from 2K to 16K Tokens Unlocks Higher Image Resolution



RoPE Base Adjustment

Increased from 10K to 273K for extended context



Mixed Context Fine-tuning

Training on long and short context data

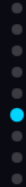


Model-specific Settings

2.2B uses 16K; smaller variants use 8K

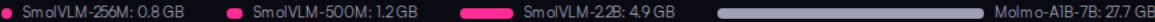


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SmoIVLM Achieves Competitive Performance at Sub-1GB to 4.9GB RAM

Capability	Benchmark	256M	500M	2.2B	Best Efficient OS
Single-Image	OCRBench	52.6%	61.0%	72.9%	54.7%
	AI2D	46.4%	59.2%	70.0%	71.0%
	ChartQA	55.6%	62.8%	68.7%	48.0%
	TextVQA	50.2%	60.2%	73.0%	61.5%
	DocVQA	58.3%	70.5%	80.0%	77.7%
	ScienceQA	73.8%	80.0%	89.6%	87.5%
Multi-task	MMMU	29.0%	33.7%	42.0%	33.9%
	MathVista	35.9%	40.1%	51.5%	37.6%
	MMSStar	34.6%	38.3%	46.0%	43.1%
Video	Video-MME	33.7%	42.2%	52.1%	45.0%
	MLVU	40.6%	47.3%	55.2%	48.2%
Average (All 11 Benchmarks)		44.0%	51.0%	59.8%	—



Real-Time Inference on iPhones: Up to 80 Tokens/Second on Apple Silicon



100% On-Device
HuggingSnap processes photos and videos locally — no cloud dependency

Key Takeaways: Design Principles for Efficient Multimodal Models

- 1 Balance encoder-LM compute:** Smaller encoders outperform larger ones for compact VLMs — allocate parameters wisely.
- 2 Extend context length:** RoPE base adjustment from 10K to 273K enables 16K token contexts for higher resolution images.
- 3 Use aggressive token compression:** $r=4$ pixel shuffle provides 16× reduction, critical for small model efficiency.
- 4 Sub-1GB inference is achievable:** SmoIVLM-256M at 0.8GB surpasses the 300× larger LlaVA-768.
- 5 Fully open-source:** All weights, datasets, code, and HuggingSnap app enable reproducibility.

Architecture efficiency matters more than raw scale

SmoIVLM: From 256M to 2.2B parameters — all deployable on edge devices